

ONION PRODUCTION: A PRACTICAL GUIDE FOR SMALLHOLDER FARMERS IN ETHIOPIA



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1. INTRODUCTION

Common Name: Onion

Scientific name: *Allium cepa*

Local name: ቀይ ሸንኩርት (Amharic)

Onion is a member of the family Amaryllidaceae and one of the most widely cultivated species of the genus *Allium*. Onion is a popular vegetable grown in Ethiopia mostly for its pungent bulbs, although the leaves are also edible in other countries. The bulb is composed of concentric, fleshy, enlarged leaf bases or scales. The outer leaf bases lose moisture and become scaly and the inner leaves generally thicken and develop as bulbs. The green leaves above the bulb are hollow and arise sequentially from the meristem at the innermost point at the base of the bulb. The stem is very small and insignificant during vegetative growth. The onion root system is fibrous, spreading just beneath the soil surface to a distance of 30 to 46 cm. Generally, onion has weak and few lateral roots.

A single terminal umbel produces numerous flowers (Fig. 1, left). The number of individual flowers per umbel varies considerable (from several hundred to over thousand). Onion is insect-pollinated (cross pollinated). The seeds are contained in a black, three-celled capsule. Usually two seeds are born in each locule of the capsule (Fig. 1, right).



Fig. 1: Flowering onion plants (left) and onion seeds (right)

The onions can be divided in to three groups depending to their growth characteristics, yet all belonging to the species *Allium cepa*.

1. Common onion • Bulb formed on a single plant • Inflorescence does not form bulbets • Commercially important • Propagated by true seed	
2. Aggregatum group • A single plant forms cluster of several bulbs • The bulbs are smaller than those of common onions • The inflorescence does not form bulbets • The group is propagated by vegetative means (daughter bulbs) • E.g. potato or multiplier onion, and shallots	
3. Proliferous group • Ground bulbs are poorly developed • Plants produce a cluster of bulblets where a normal onion would have flowers • They are less important in commercial production • Their common name are tree onion, top-setting onions, or Egyptian onion	

Onion bulb is an important spice for foods including salad, soups and stews. It is rich in Calcium, Iron, Vitamin B, and Vitamin E and has therapeutic properties.

Onion is cold season crop and preferably grown in mid altitude of Ethiopia. It is one of the most important vegetables grown as a source of income for smallholder farmers throughout

the country. The annual production of onion in Ethiopia during the 2021/2022 production season was estimated to be 268,737.43 tons produced on 31,579 hectares of land with the productivity of 8.5 t/ha (FAO, 2022). The production and productivity of onion in Ethiopia is generally very low compared to other African countries. For instance, the productivity of onion in Kenya is about 15.3 t/ha (FAO, 2022).

In areas with heavy rainfall and humid climate, onion is mainly produced during the off-season under irrigation production system in Ethiopia while in low humid area onion can be also produced during the rainy season using intensive fungicide applications. Farmers in the country are using open-pollinated onion varieties with low yielding potentials and implement improper agronomic practices. Nowadays however, hybrid onion varieties with yielding potential are introduced in the country. This production guideline is therefore developed for smallholder-onion growers with the purpose of describing the principles and procedures of onion production by considering the hybrid onion varieties.

2. CLIMATIC AND SOIL REQUIREMENTS

The performance of onion depends on the cultivar and environmental conditions. Environmental conditions affect the foliage size and color, flavor (pungency) and storage quality of the bulb. Onion can be grown up to 2,200 m above sea level. Onions require well-distributed rainfall of between 350 and 650 mm during the growing period. In area with high natural rainfall or lack of adequate evapo-transpiration, the maturation period may be prolonged. High rain fall may cause thickening of the neck region of onion that cannot dry out properly. Bulbs with neck-thicken are not storable. Dry spell is needed at maturity.

The optimum temperature for onion growth is 18 to 24 °C day temperature and 10 to 15 °C night temperature. If the temperature exceeds 30 °C, maturity is hastened and small bulbs are produced, consequently lowering the yields. Higher temperatures (25-27 °C) speed up bulbing. Low temperatures (8 – 13 °C) trigger bolting (flowering) of plants.

Onions require fertile and well-drained soil. The optimum pH range is 6.0 - 6.8. Soils between sandy and silty-loam in texture with a fine-tillth are appropriate for onion production.

3. CHOICE OF VARIETY AND SEEDLING PRODUCTION

3.1. Choice of Onion Variety

The choice of adaptive and high yielder variety is very critical for good yields and successful production of onions. The choice of varieties should be based on their availability and adaptability to the growing conditions, resistance/tolerance to diseases and insect pests and yielding potentials.

The Ethiopian Institute of Agricultural Research (EIAR) and Regional Agricultural Research Institutes released various open-pollinated varieties of onion (Table 1). However, Bombay Red and Adama Red varieties are dominantly grown by the smallholder farmers. Recently however, hybrid onion varieties are registered by different seed companies (Table 2) and produced by smallholder farmers in Ethiopia. Hybrids are often high yielder and their seeds are usually more expensive than inbred/ open-pollinated onion varieties. The seeds of hybrids cannot be harvested and saved for future planting.

Table 1: Selected open-pollinated onion varieties grown in Ethiopia

Variety	Altitude	Maturity days	Yield (t/ha)	Seed yield ability	Bulb color
Adama Red	700-2000	110-130	30-35	high	Black red
Bombay Red	-	135-145	30.0	high	Light red
Melkam	1100-1800	120-150	40	high	red
Nasik red	500-1900	100-115	26-30	-	Medium red
Red Creole	-	130-140	27-30	Difficult	Light red
Nafis	-	90-110	40	High	Medium red

Source: Compiled from Megersa (2017) and Zeleke and Derso (2015)

Table 2: Selected hybrid onion varieties registered by private companies in Ethiopia

Variety	Year of registration	Registered company
Neptune	2009	Green Life
Sivan	2011	Green Life
Russet	2012	Green Life
Ada	2013	Green Life
Red Coach		Green Life
Red King	2011	Markos PLC
Jambar	2011	Makubo Enterprise PLC
Red Passion	2011	Crop Grow
Caramel	2012	Impact Mundial Agri. PLC

Source: Compiled from Megersa (2017) and Zeleke and Derso (2015)

3.2. Onion Seedlings Production

Onions can be produced either by direct seeding on the production field or by transplanting seedlings produced in nursery depending on the growing conditions of specific region. Producing onions by direct seeding requires precision seeders that allow exact placing of the seeds and controlling the patterns of the plant spacing. Moreover, direct seeding requires proper land leveling. In Ethiopian conditions however, onions are produced usually using seedlings produced in nursery. In this regard, farmers can either purchase the seedlings from certified nursery enterprises or produce their own seedlings in the nursery. In case of self-production, it is advised to produce seedlings on raised based using certified seeds purchased from seed companies.

3.2.1. Seedbed preparation and sowing

Onion seedlings are produced mostly on raised beds for draining excess water. Beds are mostly prepared with one meter width and 5 m length (Fig. 3). Incorporation of well-decomposed compost is advised for proper nutrition and for enhanced water holding capacity of the soil. Seeds purchased from recognized seed suppliers are drilled in rows made at 15 cm apart on the seedbeds and covered lightly with soil and mulch (e.g. straw). Water the seedbeds after mulching. While drilling, it is necessary to avoid dense sowing of seeds as it results weak plants that established poorly in the production field. Sowing density on nursery seedbed should be between 1,500 –2,500 seeds/m² about 7 g of seed/m²). The seeding rate of 3.0-4.0 kg/ha and 6.5-7.5 kg/ha is required hybrid and open-pollinated onion varieties (Ethio-SHEP, 2019).



Fig. 3: Seedbed (left) and onion seedlings (right) in nursery

3.2.2. Seedling management

Seedling management is a critical operation in onion nursery. It helps producing healthy, strong and uniform planting materials that establish easily after transplanting in the production field. Remove mulching materials as soon as seeds are germinated. Otherwise seedlings will be long, weak and thin. Irrigate the seedbeds generously in the first 7-10 days; then after light irrigation is preferable to reduce the incidence of damping-off; an important disease of seedling in the nursery. Application of pesticides could be necessary to manage diseases and insect pests in the nursery.

4. ESTABLISHMENT AND MANAGEMENT OF ONION FARM

5.1 Field Preparation

The production field of onion should be near to reliable water source as onions require frequent watering. Moreover, avoid planting of onion on fields previously planted with onion and related vegetables. Crop rotation prevents the build-up of insect pests and diseases and it replenishes soil nutritional status of the soil. A 3-4 years crop rotation program is recommended for onion (Fig. 4).

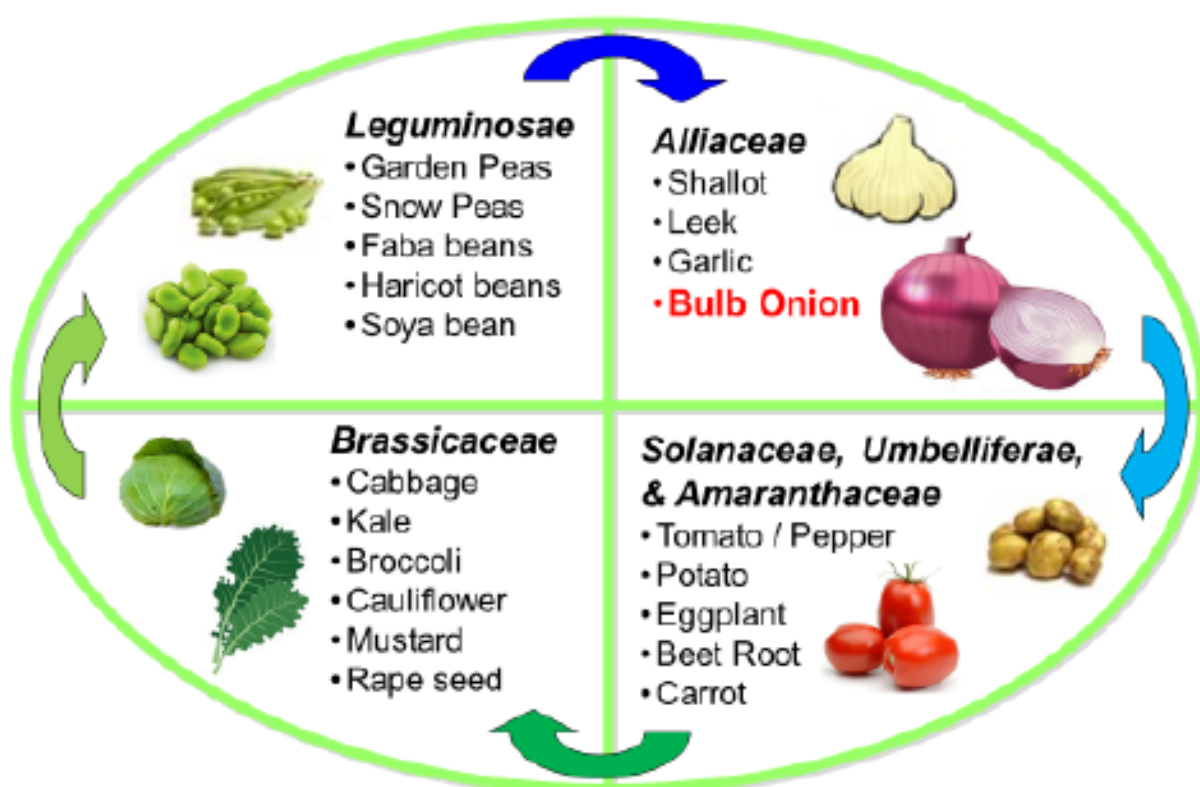


Fig. 4: Proper crop rotation schedule for onion

Source: Ethio-SHEP (2019)

Field preparation can be done by oxen-plow for small scale production (Fig. 5) or using tractors in commercial large-scale production of onions. Production fields should be ploughed and sufficiently disked ensures a fine tilth of the soil. While plowing incorporate the crop residues into 30 cm soil depth about one to two months before transplanting. After plowing level the land and apply and incorporate well-decomposed compost into the soil about 1-2 weeks before transplanting, preferably by hoeing. Compost can be made using locally available materials including manure, green leaves and branches and any biological wastes. Composts improve the fertility status and water holding capacity of the soil.



Fig. 5: Plant bed preparation using traditional oxen-plow

Source: Ethio-SHEP, (2019)

5.2 Transplanting

Transplanting is the transfer of seedlings into the production field. Before transplanting, it is important to harden the seedlings in nursery for enhanced establishment of the seedlings in the production field. It is done by slightly reducing the water supply, removing the shade (if any) and exposing the seedlings to strong sunlight so that they will be stocked and strong. Hardening should be started about ten days before the date of transplanting. Onion seedlings are ready for transplanting about 6-8 weeks after sowing depending on the management in the nursery. Watering the seedlings 12-14 hours before transplanting is necessary to reduce root damages while uprooting.

Onion is planted at 40 cm between ridges (double rows), 20 cm between rows and 5 cm between plants in the row (Fig. 6) as indicated by Ethio-SHEP (2019). It is advised to irrigate the field before transplanting. Transplanting the seedlings late in afternoon or cloudy days

minimizes the transplanting shock of seedlings. Immediately after transplanting it is necessary to irrigate seedlings as it enhances the seedling establishment rates.



Fig. 6: Spacing of onion plants in the production field

Source: Ethio-SHEP (2019)

5.3 Watering

Onion requires generally light and frequent watering/irrigation; because onion plants have shallow root system (about 30 cm deep). Excessive irrigation at early growth stage should be avoided, as it increases the disease incidence and retards plant growth. At bulbing stage onion requires substantial amount of water while towards maturity the irrigation water should be reduced/stopped.

In case of light soils, frequent application of less water is preferable while the amount of water could be increased as the plant and the roots increase in size. About 3-4 weeks before harvesting irrigation water should be stopped to facilitate onion neck drying and bulb curing. Furrow irrigation is general proffered and practiced in Ethiopia for onion production (Fig. 7).



Fig. 7: Furrow irrigation system in onion farm

Source: Ethio-SHEP (2019)

5.4 Nutrient Management

Associated with its shallow root system, onion has relatively high nutrient requirements. The application of organic (compost) and inorganic fertilizers improves the growth and bulb yield of onion by supplying the necessary nutrients and improving the structure and water holding capacity of the soil.

The quantity of fertilizers to be applied depends on the type and fertility status of the soil and the yielding potential of the variety. The rate for a particular locality or soil types has to be determined by soil analysis before application. As farmer's level soil analysis is difficult, apply 242 kg/ha NPS at the time of transplanting and urea at the rate of 100 kg/ha as top dressing in split application (Zeleeke and Derso, 2015). The first half of urea is applied about 15 to 20 days after transplanting while the second half is applied 30 to 40 days after transplanting as top dressing (Fig. 8). Generally, application of urea should be completed before the initiation of bulb. Delayed application results in thick necks and thus reduces storability of the onion bulbs.



Fig. 8: Top dressing of urea fertilizer

Source: Ethio-SHEP (2019)

5.5 Cultivation/Weeding

Hoeing is an important operation in onion production for managing weeds. Weeding also helps destroying alternate hosts of insect pests and diseases of onion. Cultivation/hoeing loosen hard soil around plants for good aeration and expansion of bulbs. It is done when the soil is moist, but not wet.

5.6 Insect Pest and Disease Management

Onion is attacked by the wide ranges of insect pests and diseases. In addition, onion is liable for different physiological disorders that are associated with adverse environmental conditions and nutrient deficiencies. For successful production, onion pests should be managed by integration of different management options including cultural, mechanical, physical, chemical and biological methods.

Major diseases, insect pests and physiological disorders that commonly attacked onion in Ethiopian conditions are described below. The symptoms and possible management options of these pests are also presented below in Table 2.

Table 2: Major diseases and insect pests of onion in Ethiopia

I. Common insect pests of onion				
Name	Description	Damages/symptoms	Damage/symptom picture	Management options
1. Onion Thrips (<i>Thrips tabaci</i> Lindeman)	Thrips prefer tight space and cause severe damages on plants with tightly packed leaves. They feed on the base of the youngest leaves and transmit viral diseases.	- Feeding causes whitish blotches that appear as silvery streaking on leaves - As feeding continues, affected tissue may turn dry and yellow, and may eventually brown and die - Black spots (insect excreta) on the silvery attacked leaves	 <i>Thrips on onion leaves</i> Source: Lowenstein and Groves (2020)	- Remove plant debris and weeds from the field - Spray Diazinon 60% EC and Daimethiot 49% EC - Spray directly into the center of the plants - Use insecticides alternately to minimize resistant thrips
2. Onion Flies	It is not as common as onion Thrips in Ethiopia. The larvae stage (maggot) feed on the root system of onion plants	- Pests eat the lateral roots causing tunnels into the taproot - Plants become shriveled or eventually die - They also feed on bulbs and the feeding damages can be entry point for secondary infection (e.g. Bacterial Soft Rot)	 <i>Onion roots and bulbs damaged by onion flies</i> Source: Ethio-SHEP (2019)	- Use well decomposed manure/compost - Practicing well planned crop rotation - Remove and destroy infested plants - Incorporate the crop residues immediately after harvest by plowing
II. Common diseases of onion				

1. Damping-off	Onion Damping-off is caused by the fungus <i>Fusarium</i>. It is seed born. The fungus survives in soil and disease emergence is favored by moist to wet soil.	• Infected seeds rot and fail to produce a seedling • Rotting seeds are covered by mold • Discolored root tips which may be pink, tan, yellow, red or black • Slowly growing seedlings that may wilt and die.	 Wilting of onion plants caused by <i>Fusarium</i> damping-off	• Use disease-free seed • Treat seeds with fungicides • Crop rotation with cereals or grasses (3-4 years) • Fumigate or solarize the soil to reduce level of <i>Fusarium</i> in the soil
2. Onion Downy Mildew	A serious disease of that prevails in cool, humid and poor drained soils. It is caused by a fungal pathogen <i>Peronospora destructor</i>.	• Pale spots or elongated patches on leaves; • Gray-purple fuzzy growth on leaf surface; • Leaves turning pale then yellow; • Collapse of leaf tips	 Fuzzy growth and collapse of leaf tip by Downy Mildew	• Use clean bulbs for planting • Crop rotation for 3-4 years • Use well-drained area • Destroy all infected crop debris; • Spray Metalaxyl + Mancozeb, Ridomil Gold, and etc.

3. Purple blotch	Caused by a fungal pathogen <i>Alternaria solani</i>. Continues rainfall or heavy dew followed by bright sunny days favors the disease. Infect all aboveground parts and the bulb. It also attacks, garlic and other allium crops.	• Small water-soaked lesions on leaves or stalk with white centers, mostly on older leaves. • Lesions enlarge and become purplish with light yellow rings on the margins. • Younger leaves more susceptible as the bulb matures. • Bulbs infected at the neck.	 Onion leaves and bulbneck infected by purple blotch fungi	• Use disease free onion seeds • Crop rotation • Practice field sanitation • Weed control • Use resistance varieties • Spray Ridomil Gold, Diprocon 33 EC, etc.
4. Bolting	It is a premature flowering of onion plants due to a sudden change in temperature (below 10 °C).	• Rapid wilt of the plants and buildup of adventitious roots on the infected stem. It causes browning of the vascular bundles. The infected stem (xylem) in water releases a slimy stream of whitish bacterial cells.	 Bolted onion plants	• Plant onions at correct planting time • Use variety with less vulnerability to bolting • Practice proper fertilization (N) and Irrigation • Snip off the flowers or flower stalks

5. HARVESTING AND POSTHARVEST HANDLING OF ONION

5.1. Harvesting

Onions are ready for harvest when the leaves collapse. Onion tops should be mechanically toppled to dry out the tops before harvesting, if natural drying is not possible. Onion bulbs are harvested when at least 75% or more of the tops have been fallen down on the ground. In irrigated onion production, irrigation should be stopped a week to ten days prior to harvesting.

Harvesting is done by undercutting the roots or digging out the bulbs using hand tools. Harvested bulbs are then windrowed in to a single row in the field to permit the drying of onion tops and the outer scale of the onion bulbs. Normally, onions are harvested when no rain is likely to occur for a few days.

5.2. Onion Curing, Yield and Storage

Curing is the process of drying the onion tops and the outer scale of the onion bulbs. Curing helps enhance the storage life and quality, protect against damages and diseases, and promotes natural dormancy. It can be done by spreading harvested bulbs in windrow in the production field under natural conditions for a week (Fig. 9, left). If rainfall is likely to occur, the harvested bulbs should be collected from the production field and cured in storage by exposure them to temperatures up to 35° in low (less than 50 per cent) relative humidity (Fig. 9, right). After curing, the dried bulbs are then topped mechanically by simultaneous squeezing and pulling the dried tops from the bulb.



Fig. 9: Curing of onion bulbs in the field under natural conditions (left) in storage (right)

The yield of onion varies with the suitability of the environmental conditions, the potentials of the varieties used, the agronomic practices implemented and the extent of losses. According to Yeshiwas et al. (2024) the productivity of onion in the last 20 years (2002 – 2021) is in decreasing trend in Ethiopia while the area coverage increased. The average national productivity of onion in Ethiopia is 8.2 t ha⁻¹ (CSA, 2022), which is very low. The low productivity is mainly associated with inappropriate agronomic practices, lack of high yielder storable varieties, perishability nature of the crop and poor market linkage (Yeshiwas et al., 2023). On the other hand, the productivity of onion can be reached up to 30.0 t ha⁻¹ using hybrid varieties (unpublished). Implementing appropriate agronomic practices and using adaptable hybrid varieties is therefore critical for the enhanced productivity of onion.

Onion is a perishable crop. Postharvest loss is a serious problem in onion production. It occurs in the value chain of the crop. According to Yeshiwas et al. (2023), the average postharvest loss of onion along the value chain is about 29.8%. Of which the major proportion of losses were observed at the farmer level of the value chain.

Onion is normally stored in temperature-humidity control storage in bulk at the temperature 0 °C and relative humidity of 60-65%. At farmer's level, onion can be stored in simple ventilated storage constructed from locally available materials such as woods and grasses or metal sheet as covering materials. Onion bulbs are then placed on shelves constructed from timber or woods (Fig. 10). Properly cured onion bulbs could be stored for 2-3 months.



Fig. 10: Storage constructed from local materials and onion stored in shelf

6. COST AND INCOME ANALYSIS OF ONION PRODUCTION

Record keeping is very important for evaluating the profitability of a given farm including onion production. Therefore, the farmers are encouraged to keep accurate records of all farm activities that incur costs. The record should include both the cost of production and sales so that the farmers are able to analyze whether onion-farming makes profit or loss.

A simplified sample sheet is presented below that helps recoding the gross income, production and sale costs and the gross benefit of onion farming.

Table 3: Major items and activities to be considered for determining gross margin of onion production

Item	Quantity	Unit price	Total	Remark
A. Input costs				
Seed				
Manure/compost				
Fertilizer				
Staking wood				
Chemicals				
B. Costs for land preparation				
C. Labor cost for planting				
D. Management costs				
Cultivation				
Watering and fertilizer application				
Harvesting				
E. Transportation and packaging cost				
F. Total cost (A+B+C+D+E)				
G. Marketable yield				Gross income
H. Net income (G-F)				

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